Biology

Topic: Cell cycle cell division

Cell Cycle

1. Consider the following statements related to interphase and select the correct option

(A) Is a biosynthetic phase in which the cell duplicates its organelles.

(B) Constitutes more than 95% duration of cell cycle.

(C) Is the phase between two successive M-phases.

1) Only (A) is correct
2) Only (B) and (C) are correct
3) Only (C) is incorrect
4) All (A), (B) and (C) are correct

2. The second gap phase of interphase present between S and M phase includes all, except

1) Duplication of mitochondria, chloroplast, Golgi bodies
2) Centriole duplication
3) Synthesis of tubulin protein
4) Synthesis of RNA

3. Consider the following statements and select the correct option A, B and C

(i) The differentiation of the cells occurs in ___ A ___ stage of the cell cycle.

(ii) DNA replication occurs in ___ B ____ phase of cell cycle.

(iii) Proteins for spindle fibre formation are synthesised in __ C ___ phase

	A	B	C
1)	G_0	M	G_1
2)	G_0	S	G_2
3)	M	G_0	G_2
4)	M	G_2	G_1

4. Match the following columns and select the correct option:

Column I		Column II	
1)	Interphase	(i)	Cell quiescent
2)	M-phase	(ii)	Longest phase of the cell cycle
3)	S-phase	(iii)	Less than 5% of total duration of cell cycle
4)	G_0-phase	(iv)	DNA content doubles

1) A– (ii), B - (iii), C - (iv), D- (i)
2) A- (ii), B – (ii), C -(i), D- (iv)
3) A-(iii), B – (ii), C- (iv), D - (i)
4) A- (iii), B - (i), C– (iv), D – (ii)

5. G_0 stage of cells in eukaryotic cell cycle denotes

1) Check point before entering the next phase

2) Pushing in the middle of a cycle to cope with a temporary delay
3) Death of a cell
4) Exit of cells from cell cycle

6. Interphase of the cell cycle is called the resting phase because
1) It is inactive phase of the cell cycle
2) It does not involve any activity related to cell division
3) It does not prepare cell for cell division
4) It is the phase where cell rests before entering into mitosis

7. In an onion root tip cell, there are 24 chromosomes. How many chromosomes will be present in the meristematic cells of onion
(A) In G_1 phase
(B) After S phase
(C) After M phase

	A	**B**	**C**
1)	24	24	24
2)	24	24	48
3)	12	24	48
4)	24	48	12

8. Match the following columns and choose the correct option.

Column I		Column II	
1)	Interphase	a	Separation of daughter chromosomes
2)	Karyokinesis	b	Actual mitosis occurs
3)	M phase	c	Resting phase
4)	Cytokinesis	d	Division of cytoplasm

1) 1-c, 2-a, 3-b, 4-d
2) 1-b, 2-a, 3-c, 4-d
3) 1-c, 2-d, 3-b, 4-a
4) 1-b, 2-d, 3-c, 4-a

9. During _____ phase, proteins are synthesised in preparation for M phase.
1) G_1 phase
2) Go phase
3) G_2 phase
4) S phase

10. Read the following statements and choose the correct option
(i) DNA synthesis occurs only during one specific stage in the cell cycle.
(ii) All cells reproduce by dividing into two, with each parental cell giving rise to two daughter cells.
(iii) A typical eukaryotic cell cycle is illustrated by human cells in culture.
(iv) During the division of a cell, DNA replication and cell growth also takes place in an orderly manner.
1) All statements are correct except (i)
2) All statements are correct except (iv)
3) All statements are correct

4) All statements are correct except (iii)

11. Read the following statements w.r.t.cell cycle:

(A) M-phase is also called the resting phase because replication does not occur during this phase.

(B) Interphase is the phase during which the cell is preparing for division.

(C) The interphase is divided into four prominent phases, i.e., G_0 G_1 S and G_2 phases.

(D) Interphase represents the phase between the two successive M phases.

The option with incorrect statement is

1) (A) and (B)
2) (A) and (C)
3) (B) and (D)
4) (C) and (D)

12. Refer to the given assertion and reason carefully and select the correct option.

Assertion: If the cell had diploid or 2n number of chromosomes at G_1 phase, even after S phase the number of chromosomes remains the same, i.e., 2n.

Reason: There is no increase in the chromosome number in S phase.

1) If both assertion and reason are true and reason is the correct explanation of the assertion
2) If both assertion and reason are true but reason is not the correct explanation of the assertion
3) If assertion is true but reason is false
4) If both assertion and reason are false

13. Read the following statements and choose the incorrect option.

1) During the G_0 phase, proteins are synthesised in preparation for mitosis while cell growth continues.
2) The duration of cell cycle can vary from organism to organism and from cell type to cell type.
3) The cell cycle is divided into two basic phases.
4) These newly formed daughter cells after each mitotic division can themselves grow and divide, giving rise to a new cell population that is formed by the growth and division of a single parental cell and its progeny.

14. The duration of cell cycle in human cell is

1) 48 hours
2) 24 hours
3) 72 hours
4) 108 hours

M Phase and Significance of Mitosis

15. All statements are incorrect wr.t. mitosis except

1) Prophase is the best stage to study shape of chromosomes.

2) During anaphase, chromosomes lose their individuality.
3) Colchicine inhibits mitosis by inhibiting formation of microtubules.
4) It is also called heterotypic division.

16. Consider the following statements and select the option that correctly fills the blank:

(i) The disintegration of the nuclear membrane marks the beginning of (A).
(ii) The (B) ends when the sister chromatids reach the respective poles.
(iii) (C) marks the end of karyokinesis.

	A	**B**	**C**
1)	Metaphase	Anaphase	Telophase
2)	Prophase	Anaphase	Metaphase
3)	Prophase	Metaphase	Anaphase
4)	Metaphase	Telophase	Prophase

17. Read the following statements, and select the correct option:

Statement (A): Complete disintegration of nuclear envelope marks the start of the second phase of mitosis.

Statement (B): Metaphase chromosome is made up of two sister chromatids.

1) Only (A) is correct
2) Only (B) is correct
3) Both (A) and (B) are correct
4) Both (A) and (B) are incorrect

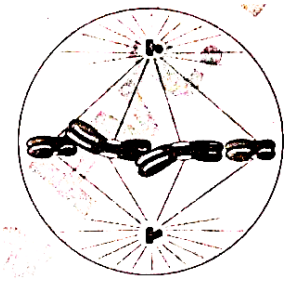
18. All are correct w.r.t. significance of mitosis except

1) Growth of an organism
2) To repair damage and to regenerate lost body parts
3) Gametogenesis in diploid organisms
4) Heal a wound

19. For the cytokinesis in plant cell, which of the following cell organelle play important role?

1) Lysosome
2) Endoplasmic reticulum
3) Dictyosomes
4) Cytoskeletal system

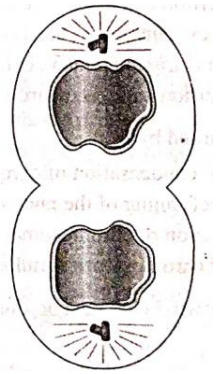
20. Mark the option which truly represents the event of cell division stage in given diagram.



1) Splitting of centromere
2) Spindle fibre attached to kinetochore of chromosomes

3) Nucleus, Golgi apparatus, ER reform
4) Sister chromatids move to opposite poles

21. Identify the following stage of cell division. Choose the correct option regarding this stage.



1) Nucleolus, Golgi apparatus, and ER reform.
2) The chromatin material tends to collect at each of the two poles.
3) Chromosome identity is lost as discrete elements.
4) All of these.

22. Read the following statements w.r.t. M-phase and select the correct option:

(A) Mitosis involves a series of changes in the nucleus only.
(B) Clear cut lines cannot be drawn between various stages.
(C) Cytokinesis occurs after karyokinesis.
(D) In M-phase, cell duplicates its cellular components and distributes them in random manner.

1) (A), (B) and (C) are correct
2) (B) and (C) are correct
3) (B) (C) and (D) are correct
4) (A), (C) and (D) are correct

23. Which of the events listed below is not observed during mitosis?

1) Chromatin condensation
2) Movement of centrioles to opposite poles
3) Appearance of chromosomes with two chromatids joined together at the centromere
4) Crossing over

24. Read the given statements of assertion and reason and select the correct option.

Assertion: By the end of prophase stage, condensation of chromosomes is completed, hence they can be observed clearly under the microscope.

Reason: Metaphase chromosome is made up of two sister chromatids, which are held together by the kinetochores.

1) If both assertion and reason are true and reason is the correct explanation of the assertion
2) If both assertion and reason are true but reason is not the correct explanation of the assertion
3) If assertion is true but reason is false

4) If both assertion and reason are false

25. Prophase is marked by

1) Initiation of condensation of chromosomal material
2) Complete reforming of the nuclear envelope
3) Decondensation of chromosomal material
4) Splitting of chromosomes simultaneously

26. The events shown below occur during which of the phases?

(A) Chromosome identity is lost as discrete elements.

(B) Condensation of chromosome is completed, and they can be observed clearly under the microscope.

(C) Nuclear envelop disintegrates

(D) Centromere split and chromatids separate.

	A	**B**	**C**	**D**
1)	Telophase	Prophase	Metaphase	Anaphase
2)	Telophase	Metaphase	Prophase	Anaphase
3)	Anaphase	Prophase	Telophase	Metaphase
4)	Prophase	Metaphase	Anaphase	Telophase

27. Read the given statements of assertion and reason and select the correct option.

Assertion: Plant cells undergo cytokinesis by a different mechanism than animal cells.

Reason: Plant cells are enclosed by a relatively inextensible cell wall.

1) If both assertion and reason are true and reason is the correct explanation of the assertion
2) If both assertion and reason are true but reason is not the correct explanation of the assertion
3) If assertion is true but reason is false
4) If both assertion and reason are false

28. A very significant contribution of mitosis is

1) Gametogenesis in diploid organisms
2) Cell repair
3) Sperm and ova formation in humans
4) All of the above

29. Read the following statements and choose the incorrect option.

1) At the onset of anaphase, each chromosome arranged at the metaphase plate is split simultaneously.
2) The chromosomal material becomes untangled during the process of chromatin condensation.
3) At theend of metaphase, the two daughter chromatids, now referred to as daughter chromosomes of the future daughter nuclei, move to opposite poles.
4) At the end of telophase, chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.

30. Mitosis helps to

1) Restore nucleocytoplasmic ratio of cell

2) Heal a wound
3) Produce gametes in sexually reproducing organism
4) Both (1) and (2)

Meiosis and Significance of Meiosis

31. Anaphase of the mitosis is different from anaphase I, as it shows
1) Alignment of chromosomes at the equator
2) Separation of homologous chromosomes towards the opposite poles
3) Splitting of centromere and separation of sister chromatids towards the opposite poles
4) Attachment of spindle fibre from opposite poles to the kinetochores of sister chromatids

32. Beginning of diplotene stage is characterised by
1) Pairing of homologous chromosomes
2) Exchange of genetic material between non-sister chromatids of homologous chromosomes
3) Dissolution of synaptonemal complex
4) Terminalisation of chiasmata

33. Identify the meiotic stages in which given events occur.
(A) The chromosome becomes gradually visible with compaction of chromatin.
(B) The complex is formed by a pair of synapsed homologous chromosomes.
(C) The four chromatids of each bivalent chromosome becomes distinct and clearly appear as tetrad.

	A	B	C
1)	Leptotene	Pachytene	Diplotene
2)	Leptotene	Zygotene	Pachytene
3)	Zygotene	Pachytene	Diplotene
4)	Zygotene	Diplotene	Pachytene

34. Choose the odd one w.r.t. significance of meiosis:
1) Helps in maintaining the chromosome number of each species.
2) Increases genetic variability.
3) Introduces new combination of traits.
4) Replacement of cells of lining of gut of human body.

35. Meiosis in diploid organisms results in
1) Production of gametes
2) Reduction in the number of chromosomes
3) Introduction of variation
4) All ofthe above

36. In oocytes of some vertebrates ____ can last for months oryears.
1) Leptotene
2) Zygotene
3) Diplotene

4) Diakinesis

37. Read the following statements:

(A) Terminalisation of chiasmata occurs

(B) Fully condensed chromosomes

(C) Meiotic spindle is assembled

(D) Represents transition to metaphase

The above characteristics are attributed to

1) Zygotene
2) Pachytene
3) Leptotene
4) Diakinesis

38. Match the given columns and choose the correct option.

Column I		Column II	
1)	Zygotene	(i)	Formation of recombination nodule
2)	Diakinesis	(ii)	Synapsis
3)	Diplotene	(iii)	Dissolution of synaptonemal complex
4)	Pachytene	(iv)	Terminalisation of chiasmata

1) A-(ii), B-(iv), C- (iii), D- (i)
2) A– (ii), B — (iii), C – (iv), D – (i)
3) A-(i), B- (iii), C – (iv), D – (ii)
4) A-(i), B- (iv), C – (ii) D - (iii)

39. Select the mismatched pair:

1) Diplotene: Can last for month or years in some vertebrate oocytes
2) Meiosis: Increases genetic variability in a population of organisms
3) Pachytene: Synaptonemal complex dissolves and chiasmata are seen in the form of X-shaped structures
4) Cytokinesis: Achieved by formation of cell plate that represents the middle lamella in plant cells

40. In meiosis I, a bivalent is an association of

1) Four chromatids and four centromeres
2) Two chromatids and two centromeres
3) Two chromatids and one centromere
4) Four chromatids and two centromeres

41. Select the incorrect statement.

1) Meiosis involves 2 sequential cycles of nuclear and cell division
2) Meiosis involves only single cycle of DNA replication
3) Meiosis involves pairing of homologous chromosomes
4) Meiosis involves recombination between sister chromatids of non-homologous chromosomes.

42. Arrange the following events of meiosis in a correct sequence:

(A) Terminalisation

(B) Crossing over

(C) Synapsis

(D) Disjunction of chromosomes

1) C, B, A, D
2) C, B, D, A
3) B, A, C, D
4) C, D, B, A

43. Stage between the meiosis I and meiosis II is

1) Called interkinesis
2) Characterised by polymerisation of deoxyribonucleotides
3) Short lived
4) More than one option is correct

44. In a diploid cell, chromosomes of a pair are similar in form, size and structure. Select the incorrect statements w.rt. these chromosomes:

1) These chromosomes are also called homologous chromosome.
2) During meiosis these chromosomes separate from each other.
3) They form monovalent during zygotene stage.
4) Pairing of these chromosomes is called synapsis.

45. In metaphase II of meiosis

(A) The chromosomes align themselves at the equator.

(B) The centromere holding the two chromatids aligns at equatorial plate and microtubules from the opposite pole of the spindle get attached to the kinetochores.

1) Both (A) and (B) are correct
2) Both (A) and (B) are incorrect
3) Only (A) is correct
4) Only (B) is correct

46. Metaphase I and Metaphase II are different from each other because

1) In metaphase I homologous chromosomes line up in pair along metaphasic plate in contrary to individual chromosomes lining up in metaphase II
2) Composition of spindle fibres
3) Velocity of movement of chromatids
4) All of these

47. Diplotene phase of meiosis is also characterised by

(A) Dissolution of synaptonemal complex

(B) Synapsis

(C) Appearance of X-shaped structure

(D) Complete terminalisation of chiasmata

1) (A), (C) and (D)
2) (B), (C) and (D)
3) (A) and (C)
4) (B) and (D)

48. Give the name of the phase of meiosis, in which

(A) The chromosome number is reduced to haploid state

(B) The amount of DNA is teduced to half

	A	B
1)	Anaphase I	Anaphase II
2)	Anaphase II	Anaphase I
3)	Metaphase II	Anaphase I
4)	Anaphase I	Metaphase II

49. Read the following statements and choose the correct option.

1) Meiosis II is initiated immediately after cytokinesis, usually before the chromosomes have fully elongated.
2) In contrast to meiosis. II, meiosis I resembles a normal mitosis.
3) At the end of telophase, the chromosomes again become compact.
4) All of the above

50. A synaptonemal complex is formed during __A__ stage and dissolves during __ __ B ____ stage. Complete the above statement by choosing the correct option for A and B.

	A	B
1)	Leptotene	Zygotene
2)	Zygotene	Diplotene
3)	Leptotene	Diplotene
4)	Pachytene	Diakinesis

Answer Keys

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
4	2	2	1	4	2	1	1	3	3	2	1	1	2	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
1	3	3	3	2	4	2	4	4	1	2	1	2	3	4
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
3	3	2	4	4	3	4	1	3	4	4	1	4	3	1
46	**47**	**48**	**49**	**50**										
1	3	1	1	2										